

Hanjun Park

Department of Industrial, Manufacturing, & Systems Engineering
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EDUCATION

Ph.D., Industrial and Systems Engineering 08/2025
Virginia Tech, Blacksburg, VA
Advisors: Maury A. Nussbaum and Divya Srinivasan

Graduate Certificate in Data Analytics
Virginia Tech, Blacksburg, VA

M.S., Industrial Engineering 08/2014
Seoul National University, Seoul, Korea
Advisor: Woojin Park

B.S., Industrial and Systems Engineering 12/2011
Georgia Institute of Technology, Atlanta, GA

ACADEMIC APPOINTMENTS

Associated Member, Faculty of Bioengineering 04/2026–Present
Texas Tech University

Assistant Professor 09/2025–Present
Texas Tech University

- Department of Industrial, Manufacturing, and Systems Engineering

PUBLICATIONS

Peer-Reviewed Journal Publications (* corresponding-author)

- [J.9] Santos, F., Son, C.*, **Park, H.**, Xue, C., Yang, J., & Liang, N. (2026). Biomechanical effects of virtual reality headset support and task demands on the neck. *IIE Transactions on Occupational Ergonomics and Human Factors*, 1-19. doi:10.1080/24725838.2026.2680465
- [J.8] Santos, F., Son, C.*, **Park, H.**, Xue, C., Yang, J., & Liang, N. (2026). Design, fabrication, and testing of an ergonomic intervention for a virtual reality headset using 3D printing. *Ergonomics*, 1-15. doi:10.1080/00140139.2026.2653007
- [J.7] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2026). Adaptation to a whole-body powered exoskeleton: Human-exoskeleton coordination during load-handling tasks. *Annals of Biomedical Engineering*, 1-15. doi:10.1007/s10439-026-04025-9

- [J.6] **Park, H.***, & Nussbaum, M.A. (2025). Muscle synergy analysis during pseudo-static and dynamic overhead tasks with arm-support exoskeletons. *Journal of Biomechanics*, 113135. doi:10.1016/j.jbiomech.2025.113135
- [J.5] Ransing, V., Park, J., Ye, Y., **Park, H.**, Kim, S., Du, J., & Srinivasan, D.* (2025). Efficacy of virtual reality-based approach for users to understand the potential benefits and limitations of using exoskeletons. *Human Factors*, 67(11), 1136-1151. doi:10.1177/00187208251346627
- [J.4] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A.* (2025). Passive arm-support and back-support exoskeletons have distinct phase-dependent effects on physical demands during cart pushing and pulling: An exploratory study. *Applied Ergonomics*, 126, 104510. doi:10.1016/j.apergo.2025.104510
- [J.3] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2023). A pilot study investigating motor adaptations when learning to walk with a whole-body powered exoskeleton. *Journal of Electromyography and Kinesiology*, 69, 102755. doi:10.1016/j.jelekin.2023.102755
- [J.2] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2022). Effects of using a whole-body powered exoskeleton during simulated occupational load-handling tasks: A pilot study. *Applied Ergonomics*, 98, 103589. doi:10.1016/j.apergo.2021.103589
- [J.1] **Park, H.**, Park, W.*, & Kim, Y. (2014). Manikin families representing obese airline passengers in the U.S. *Journal of Healthcare Engineering*, 5(4), 479-504. doi:10.1260/2040-2295.5.4.479

Manuscripts Under Review

- [R.6] **Park, H.***, & Nussbaum, M.A. Muscle activity and joint kinematic adaptations when using passive and active back-support exoskeletons across multiple sessions of repeated symmetric lifting and lowering. *Submitted*
- [R.5] Jeong, Y., Lee, J.H., **Park, H.**, Choi, W.Y.*, Hwang, D.* How Motion Parameters Shape Perceived Usability of an Autonomous Wheelchair: Age-Related Differences Across Deceleration and Speed Modes. *Submitted*
- [R.4] Choi, B., Park, J.*, & **Park, H.** Evaluation of passive arm-support exoskeletons in repetitive tasks: physical demands, task performance, and usability effects by task direction. *Submitted*
- [R.3] **Park, H.***, & Nussbaum, M.A. Perceptual adaptations to back-support exoskeletons: evidence from measures of workload and discomfort. *In Revision*
- [R.2] Santos, F., Son, C.*, **Park, H.**, Xue, C. Yang, J., & Liang, N. Ergonomic Design of Head-Neck Supports Using Additive Manufacturing. *In Revision*
- [R.1] **Park, H.***, & Nussbaum, M.A. Characterizing adaptation to using exoskeletons: A narrative review and assessment framework for occupational contexts. *In Revision*

Peer-Reviewed Conference Proceedings

- [C.6] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *Proceedings of the 66th Human Factors and Ergonomics Society Annual Meeting*, 66(1), 273–274.
- [C.5] **Park, H.,** Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *Proceedings of the 65th Human Factors and Ergonomics Society Annual Meeting*, 65(1), 1367–1368.
- [C.4] **Park, H.,** Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *Proceedings of the 64th Human Factors and Ergonomics Society Annual Meeting*, 64(1), 888–889. - **Won Best Student Paper Award at the Occupational Ergonomics Technical Group**
- [C.3] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2014). A review on interaction techniques in virtual environments. *Proceedings of the 2014 International Conference on Industrial Engineering and Operations Management* (pp. 1582-1590).
- [C.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.,** Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *Proceedings of the 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*
- [C.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *Proceedings of 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

FUNDED RESEARCH PROJECTS AND GRANTS

External Research Grants

- [EG.1] **Biomechanical and Usability Assessment of Occupational Exoskeletons for Nurses in Overweight and Obese Patient Handling**
 Sponsor: Heartland Center for Occupational Health & Safety Pilot Project Award (NIOSH ERC)
 Total Amount: \$19,337
 Dates: 2026 – 2027
 Role: Principal Investigator
 Co-PI: Changwon Son

PRESENTATIONS

Conference Oral Presentations

- [O.13] **Park, H.,** & Nussbaum, M.A. (2026). Adaptation to using passive and active back-support exoskeletons during repetitive lifting tasks. *IISE Annual Conference & Expo 2026, Arlington, TX.*

- [O.12] Santos, F., & Son, C., **Park, H.**, (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *IISE Annual Conference & Expo 2026, Arlington, TX.*
- [O.11] Santos, F., & Son, C., **Park, H.**, (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *HFES Regional Meeting, College Station, TX.*
- [O.10] Santos, F., & Son, C., **Park, H.**, (2026). A 3D-printed shoulder support to reduce neck strain during virtual reality device use. *30th Applied Ergonomics Conference, Arlington, TX.*
- [O.9] Santos, F., **Park, H.**, & Son, C. (2025). A 3D-printed shoulder support to reduce neck strain during prolonged virtual reality (VR) device use: pilot findings. *69th Human Factors and Ergonomics Society Annual Meeting, Chicago, IL.*
- [O.8] **Park, H.**, Ojelade A., Kim, S., & Nussbaum, M.A. (2024). Effects of using arm-support exoskeletons on muscle coordination during a pseudo-static overhead task: preliminary analysis of muscle synergy. *68th Human Factors and Ergonomics Society Annual Meeting, Phoenix, AZ.* - **Won Best Experimental Paper Award at the Occupational Ergonomics Technical Group.**
- [O.7] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A. (2023). Effects of using passive back- and arm-support exoskeletons for cart pushing and pulling. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.6] Noll, A., Nussbaum, M.A., Kim, S., & **Park, H.** (2023). Reliability Analysis of Task Performance and Subjective Measures for Assessment of Occupational Exoskeletons. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.5] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *66th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.4] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *65th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.3] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *64th Human Factors and Ergonomics Society Annual Meeting.*
- [O.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*
- [O.1] Jung, J., **Park, H.**, Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

Poster Presentations

- [P.4] **Park, H.,** & Nussbaum, M.A. Motor Adaptation and Perceived Exertion During Repetitive Lifting With Passive and Active Occupational Back-Support Exoskeletons. (2026). *10th World Congress of Biomechanics (WCB 2026), Vancouver, Canada, July.*
- [P.3] Santos, F., Son, C., **Park, H.,** Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of an ergonomic intervention for virtual reality headset using 3D printing. (2026). *WCOE Research Day, Texas Tech University, Lubbock, TX, February.*
- [P.2] **Park, H.,** Ojelade, A., Kim, S., & Nussbaum, M. A. (2023). Effects of different arm-support exoskeletons on muscle activation patterns during a pseudo-static overhead task - preliminary analysis of muscle synergy. *37th American Society of Safety Professionals (ASSP) Region VI Professional Development Conference (PDC) in Myrtle Beach, South Carolina, September.*
- **Won Student Presentation Award**
- [P.1] **Park, H.,** Kim, S., Nussbaum, M. A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual material handling. *2020 Annual HFES/INFORMS Poster Competition, Virginia Tech, VA, October.* - **Won Student Poster Presentation Award**

Invited Seminars

- [S.3] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Psychological Sciences, Texas Tech University, October.*
- [S.2] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University, April.*
- [S.1] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial and Systems Engineering, Kennesaw State University, March.*

Invited Panel Talk

- [T.1] **Park, H.** (2025). From Student to Professor: Exploring Careers in Occupational Ergonomics. Panel Member. *Occupational Ergonomics Technical Group of HFES, September.*

HONORS & AWARDS

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| • Best Experimental Paper Award, OETG at HFES | 2024 |
| • Best Student Poster Presentation Award, ASSP Regional Conference | 2023 |
| • IISE Future Faculty Fellow (3F) Program Awardee | 2023–2024 |
| • NIOSH Training Fellow, Virginia Tech | 2023–2025 |
| - Fellowship includes monthly stipend, tuition, and required fees | |

- ISE Graduate Travel Award (\$900), Virginia Tech 2022
- Best Student Paper Award, OETG at HFES 2020
- VT HFES Student Chapter Research Competition Award 2020
- Grado Department of ISE Fellowship, Virginia Tech 2018
- Fellowship includes monthly stipend, tuition, and required fees
- Brain Korea 21 Scholarship, Seoul National University 2014
- Lecture & Research Scholarship, Seoul National University 2013

TEACHING EXPERIENCE

Instructor of Record

Texas Tech, Dept. of Industrial, Manufacturing, & Systems Engineering

- IE 5301: Advanced Industrial Ergonomics (n = 32) Fall, 2026
- IE 4301: Engineering Design for People (n = 40) Fall, 2026
- IE 5304: Biomechanics & Work Physiology (n = 11) Spring, 2026

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 4624: Work Physiology (n = 36) Fall, 2023
- ISE 3614: Human Factors Engineering and Ergonomics (taught online, n = 7) Summer, 2023

Guest Lecturer

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics Spring, 2023
“NIOSH Lifting Equation and Controls” (n = 100)
- ISE 4644: Occupational Safety and Hazards Control Fall, 2018
“Review on Hazard Analysis and Prevention” (n = 20)

Graduate Teaching Assistant

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics Spring, 2019
- ISE 4624: Work Physiology Spring, 2019
- ISE 4644: Occupational Health and Hazards Control Fall, 2018

PROFESSIONAL EXPERIENCES

- Software Engineer, Military duty** 10/2014–03/2018
Kornic Glory, Seoul, Korea
- Research Intern** 06/2011–08/2011
LG Electronics, Seoul, Korea

ADVISING & MENTORING

Dissertation/Thesis Committee Member

- Byungkyu Choi, TTU (PhD, IMSE) (2026–Present)
Chair: Hanjun Park

- Delaney Sheppard, TTU (MS, Bioengineering) (2025–Present)
Chair: Hanjun Park
- Felipe Santos, TTU (PhD, IMSE) (2025–2026) [J.9], [J.8], [O.12], [O.11], [O.10], [O.9], [U.2]
Chair: Changwon Son
- Daniel Leibman, TTU (PhD, Psychology) (2025–Present)
Chair: Heesun Choi
- Armina Rahman, TTU (PhD, IMSE) (2025–Present)
Chair: Changwon Son
- Jessica Lillian Dedeaux, TTU (PhD, IMSE) (2026–Present)
Chair: Changwon Son

Research Mentoring

Voluntary Postdoc Position

- Leonardo Wei, Texas Tech (PhD, ISE) (02/2026–03/2026)

M.S. Students

- Alex Noll, Virginia Tech (ISE) (08/2022–08/2024) [J.4], [C.8], [C.7]

Undergraduate Students

- Enriquez Bazoalto Kaylee, Virginia Tech (ISE) (05/2025-08/2025)

PROFESSIONAL DEVELOPMENT AND SERVICE ACTIVITIES

Professional Development Training

- IISE New Faculty Colloquium - IISE 2026
- IISE Future Faculty Fellow (3F) Program - IISE 2023-2024

Service to Profession

- Session Chair, IISE Annual Conference & Expo 2026, Arlington, TX 2026
- Session Chair, the 69th HFES Annual Meeting, Atlanta, GA 2025

Archival Journal

- Reviewer, *Ergonomics* 2026–Present
- Reviewer, *Journal of Biomechanics* 2026–Present
- Reviewer, *Gait and Posture* 2026–Present
- Reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 2026–Present
- Reviewer, *IEEE Transactions on Human-Machine Systems* 2025–Present
- Reviewer, *IISE Transactions on Occupational Ergonomics and Human Factors* 2023–Present
- Reviewer, *Human Factors and Ergonomics Society Proceedings* 2021–Present

Service to Institution

- Texas Tech Regional Science Fair (K-12) Special Award Judge 2026
- Texas Tech Regional Science Fair (K-12) Ribbon Judge 2026
- Dean's Representative, Dissertation Defense of Yves Valentin
(MS in Human Factors Psychology, TTU Department of Psychological Sciences) 2026

TTU Graduate Academic Advisor	2025–Present
VT Chapter of the Human Factors and Ergonomics Society (HFES)	
• Officer: <i>Webmaster</i>	2020–2022
National Biomechanics Day	
• <i>Led lab tours and biomechanics activities for local high school students</i>	2020–2025
VT Center for the Enhancement of Engineering Diversity (CEED) TechGirls Summer Camp	
• <i>Led hands-on biomechanics activities</i>	2019
VT College of Engineering Alumni Outreach Event	
• <i>Led lab tours for alumni</i>	2019

PROFESSIONAL AFFILIATIONS

Member, Institute of Industrial and Systems Engineers (IISE)	2023–Present
Member, Human Factors and Ergonomics Society (HFES)	2020–Present
Steering Committee member, K-Human Factors and Ergonomics Society (K-HFES)	2021–Present